MINISTRY OF EDUCATION AND TRAINING
NATIONAL ECONOMICS UNIVERSITY

Falcuty of Economic mathematics - DSEB program

FINAL EXAM: PROGRAMMING FOR DATA SCIENCE

Class: DSEB64 A **Intake:**

Exam room: **Exam date:**

Time: 90 minutes

test_no = 6401

Question 1.(3 points)

a) (1,5 points) Write a class called **TextAnalyzer**. This class should have the following methods:

- **__init__(self, file_name)**: Initializes the object by reading the content from the given file (encoding="utf8").
- **normalize(self)**: to converts the text to lowercase; removes extra whitespaces and letters not in [^a-zA-Z0-9\s];
- **count_words_frequency (self)**: Returns a dictionary with the key being the word and the value being the frequency.

b) (1 point) Consider the following code:

```
import random
stories = ['alice.txt','little_women.txt','moby_dick.txt','siddhartha.txt']
random.seed(test_no) #test_no is your test_no
story = random.choice(stories)
```

Using the class **TextAnalyzer** to display the number of words (Not case-sensitive), the frequency of words in **story**

c) (0.5 points) Find words in **story** that combine the letters {'s', 't', 'r', 'o', 'n', 'a', 'e', 'h'}.

Question 2.(2 points) Consider the matrix:

```
import numpy as np
np.random.seed(test_no)
A = np.random.randint(test_no, 8000,size = 64).reshape(8,8)
```

a) Calculate the sum of the even entries in A.

b) Find the entries in A that are a product of two prime numbers.

Question 3. (4 points)

a) (1 points) Read the grade data of all students from the 'grade_dseb64A.json' file in the folder Data/ and store them in variable **students_data.**

b) (1 points) Write a function called **calculate_average_grades(students_data)** to calculate the average grades for each student based on their grades in different courses. This function should return a dictionary with the key being the student's name and the value being average grades.

c) (1 points) Write a function to convert students' grades from a 100-grade system to a 4-grade system by the formula: a = (b/100)*4 where b is the grade in a 100-grade system and a is the converted result. Save the results to the file 'Grades_4.json' and remember to submit the file with your assignment.

d) (1points) Write a function called **find_top_course(students_data)** to determine the course with the highest average grade among all courses. This function should return the name of the course and its average grade.

Question 4.(1 points) Without using packages (numpy, scipy,...), write a function to calculate:

$$test_no^{test_no} \ mod \ 1234$$

----End----

Note: Students are allowed to use the materials (textbook and slides).

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Falcuty of Economic mathematics - DSEB program

FINAL EXAM: PROGRAMMING FOR DATA SCIENCE

Class: DSEB64 A **Intake:**

Exam room: **Exam date:**

Time: 90 minutes

test_no = 6402

Question 1.(3 points)

a) (1,5 points) Write a class called **TextAnalyzer**. This class should have the following methods:

- **__init__(self, file_name)**: Initializes the object by reading the content from the given file (encoding="utf8").
- **normalize(self)**: to converts the text to lowercase; removes extra whitespaces and letters not in [^a-zA-Z0-9\s];
- **count_words_frequency (self)**: Returns a dictionary with the key being the word and the value being the frequency.

b) (1 point) Consider the following code:

```
import random
stories = ['alice.txt','little_women.txt','moby_dick.txt','siddhartha.txt']
random.seed(test_no) #test_no is your test_no
story = random.choice(stories)
```

Using the class **TextAnalyzer** to display the number of words (Not case-sensitive), the frequency of words in **story**

c) (0.5 points) Find words in **story** that combine the letters {'s', 't', 'r', 'o', 'n', 'a', 'e', 'h'}.

Question 2.(2 points) Consider the matrix:

```
import numpy as np
np.random.seed(test_no)
A = np.random.randint(test_no, 8000,size = 64).reshape(8,8)
```

a) Calculate the sum of the even entries in A.

b) Find the entries in A that are a product of two prime numbers.

Question 3. (4 points)

a) (1 points) Read the grade data of all students from the 'grade_dseb64A.json' file in the folder Data/ and store them in variable **students_data.**

b) (1 points) Write a function called **calculate_average_grades(students_data)** to calculate the average grades for each student based on their grades in different courses. This function should return a dictionary with the key being the student's name and the value being average grades.

c) (1 points) Write a function to convert students' grades from a 100-grade system to a 4-grade system by the formula: a = (b/100)*4 where b is the grade in a 100-grade system and a is the converted result. Save the results to the file 'Grades_4.json' and remember to submit the file with your assignment.

d) (1points) Write a function called **find_top_course(students_data)** to determine the course with the highest average grade among all courses. This function should return the name of the course and its average grade.

Question 4.(1 points) Without using packages (numpy, scipy,...), write a function to calculate:

$$test_no^{test_no} \ mod \ 1234$$

----End----

Note: Students are allowed to use the materials (textbook and slides).

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Falcuty of Economic mathematics - DSEB program

FINAL EXAM: PROGRAMMING FOR DATA SCIENCE

Class: DSEB64 A **Intake:**

Exam room: **Exam date:**

Time: 90 minutes

test_no = 6403

Question 1.(3 points)

a) (1,5 points) Write a class called **TextAnalyzer**. This class should have the following methods:

- **__init__(self, file_name)**: Initializes the object by reading the content from the given file (encoding="utf8").
- **normalize(self)**: to converts the text to lowercase; removes extra whitespaces and letters not in [^a-zA-Z0-9\s];
- **count_words_frequency (self)**: Returns a dictionary with the key being the word and the value being the frequency.

b) (1 point) Consider the following code:

```
import random
stories = ['alice.txt','little_women.txt','moby_dick.txt','siddhartha.txt']
random.seed(test_no) #test_no is your test_no
story = random.choice(stories)
```

Using the class **TextAnalyzer** to display the number of words (Not case-sensitive), the frequency of words in **story**

c) (0.5 points) Find words in **story** that combine the letters {'s', 't', 'r', 'o', 'n', 'a', 'e', 'h'}.

Question 2.(2 points) Consider the matrix:

```
import numpy as np
np.random.seed(test_no)
A = np.random.randint(test_no, 8000,size = 64).reshape(8,8)
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a) Calculate the sum of the even entries in A.

b) Find the entries in A that are a product of two prime numbers.

Question 3. (4 points)

a) (1 points) Read the grade data of all students from the 'grade_dseb64A.json' file in the folder Data/ and store them in variable **students_data.**

b) (1 points) Write a function called **calculate_average_grades(students_data)** to calculate the average grades for each student based on their grades in different courses. This function should return a dictionary with the key being the student's name and the value being average grades.

c) (1 points) Write a function to convert students' grades from a 100-grade system to a 4-grade system by the formula: a = (b/100)*4 where b is the grade in a 100-grade system and a is the converted result. Save the results to the file 'Grades_4.json' and remember to submit the file with your assignment.

d) (1points) Write a function called **find_top_course(students_data)** to determine the course with the highest average grade among all courses. This function should return the name of the course and its average grade.

Question 4.(1 points) Without using packages (numpy, scipy,...), write a function to calculate:

$$test_no^{test_no} \ mod \ 1234$$

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Class: DSEB64 A **Intake:**

Exam room: **Exam date:**

Time: 90 minutes

test_no = 6404

Question 1.(3 points)

a) (1,5 points) Write a class called **TextAnalyzer**. This class should have the following methods:

- **__init__(self, file_name)**: Initializes the object by reading the content from the given file (encoding="utf8").
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b) (1 point) Consider the following code:

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import random
stories = ['alice.txt','little_women.txt','moby_dick.txt','siddhartha.txt']
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```

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Question 2.(2 points) Consider the matrix:

```
import numpy as np
np.random.seed(test_no)
A = np.random.randint(test_no, 8000,size = 64).reshape(8,8)
```

a) Calculate the sum of the even entries in A.

b) Find the entries in A that are a product of two prime numbers.

Question 3. (4 points)

a) (1 points) Read the grade data of all students from the 'grade_dseb64A.json' file in the folder Data/ and store them in variable **students_data.**

b) (1 points) Write a function called **calculate_average_grades(students_data)** to calculate the average grades for each student based on their grades in different courses. This function should return a dictionary with the key being the student's name and the value being average grades.

c) (1 points) Write a function to convert students' grades from a 100-grade system to a 4-grade system by the formula: a = (b/100)*4 where b is the grade in a 100-grade system and a is the converted result. Save the results to the file 'Grades_4.json' and remember to submit the file with your assignment.

d) (1points) Write a function called **find_top_course(students_data)** to determine the course with the highest average grade among all courses. This function should return the name of the course and its average grade.

Question 4.(1 points) Without using packages (numpy, scipy,...), write a function to calculate:

$$test_no^{test_no} \ mod \ 1234$$

----End----

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Falcuty of Economic mathematics - DSEB program

FINAL EXAM: PROGRAMMING FOR DATA SCIENCE

Class: DSEB64 A **Intake:**

Exam room: **Exam date:**

Time: 90 minutes

test_no = 6405

Question 1.(3 points)

a) (1,5 points) Write a class called **TextAnalyzer**. This class should have the following methods:

- **__init__(self, file_name)**: Initializes the object by reading the content from the given file (encoding="utf8").
- **normalize(self)**: to converts the text to lowercase; removes extra whitespaces and letters not in [^a-zA-Z0-9\s];
- **count_words_frequency (self)**: Returns a dictionary with the key being the word and the value being the frequency.

b) (1 point) Consider the following code:

```
import random
stories = ['alice.txt','little_women.txt','moby_dick.txt','siddhartha.txt']
random.seed(test_no) #test_no is your test_no
story = random.choice(stories)
```

Using the class **TextAnalyzer** to display the number of words (Not case-sensitive), the frequency of words in **story**

c) (0.5 points) Find words in **story** that combine the letters {'s', 't', 'r', 'o', 'n', 'a', 'e', 'h'}.

Question 2.(2 points) Consider the matrix:

```
import numpy as np
np.random.seed(test_no)
A = np.random.randint(test_no, 8000,size = 64).reshape(8,8)
```

a) Calculate the sum of the even entries in A.

b) Find the entries in A that are a product of two prime numbers.

Question 3. (4 points)

a) (1 points) Read the grade data of all students from the 'grade_dseb64A.json' file in the folder Data/ and store them in variable **students_data.**

b) (1 points) Write a function called **calculate_average_grades(students_data)** to calculate the average grades for each student based on their grades in different courses. This function should return a dictionary with the key being the student's name and the value being average grades.

c) (1 points) Write a function to convert students' grades from a 100-grade system to a 4-grade system by the formula: a = (b/100)*4 where b is the grade in a 100-grade system and a is the converted result. Save the results to the file 'Grades_4.json' and remember to submit the file with your assignment.

d) (1points) Write a function called **find_top_course(students_data)** to determine the course with the highest average grade among all courses. This function should return the name of the course and its average grade.

Question 4.(1 points) Without using packages (numpy, scipy,...), write a function to calculate:

$$test_no^{test_no} \ mod \ 1234$$

----End----

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Falcuty of Economic mathematics - DSEB program

FINAL EXAM: PROGRAMMING FOR DATA SCIENCE

Class: DSEB64 A **Intake:**

Exam room: **Exam date:**

Time: 90 minutes

test_no = 6406

Question 1.(3 points)

a) (1,5 points) Write a class called **TextAnalyzer**. This class should have the following methods:

- **__init__(self, file_name)**: Initializes the object by reading the content from the given file (encoding="utf8").
- **normalize(self)**: to converts the text to lowercase; removes extra whitespaces and letters not in [^a-zA-Z0-9\s];
- **count_words_frequency (self)**: Returns a dictionary with the key being the word and the value being the frequency.

b) (1 point) Consider the following code:

```
import random
stories = ['alice.txt','little_women.txt','moby_dick.txt','siddhartha.txt']
random.seed(test_no) #test_no is your test_no
story = random.choice(stories)
```

Using the class **TextAnalyzer** to display the number of words (Not case-sensitive), the frequency of words in **story**

c) (0.5 points) Find words in **story** that combine the letters {'s', 't', 'r', 'o', 'n', 'a', 'e', 'h'}.

Question 2.(2 points) Consider the matrix:

```
import numpy as np
np.random.seed(test_no)
A = np.random.randint(test_no, 8000,size = 64).reshape(8,8)
```

a) Calculate the sum of the even entries in A.

b) Find the entries in A that are a product of two prime numbers.

Question 3. (4 points)

a) (1 points) Read the grade data of all students from the 'grade_dseb64A.json' file in the folder Data/ and store them in variable **students_data.**

b) (1 points) Write a function called **calculate_average_grades(students_data)** to calculate the average grades for each student based on their grades in different courses. This function should return a dictionary with the key being the student's name and the value being average grades.

c) (1 points) Write a function to convert students' grades from a 100-grade system to a 4-grade system by the formula: a = (b/100)*4 where b is the grade in a 100-grade system and a is the converted result. Save the results to the file 'Grades_4.json' and remember to submit the file with your assignment.

d) (1points) Write a function called **find_top_course(students_data)** to determine the course with the highest average grade among all courses. This function should return the name of the course and its average grade.

Question 4.(1 points) Without using packages (numpy, scipy,...), write a function to calculate:

$$test_no^{test_no} \ mod \ 1234$$

----End----

Note: Students are allowed to use the materials (textbook and slides).